

Norepinephrine (noradrenaline): Drug information - UpToDate

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For additional information see "Norepinephrine (noradrenaline): Patient drug information" and "Norepinephrine (noradrenaline): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Levophed

Brand Names: Canada

- Levophed [DSC]

Pharmacologic Category

- Alpha-/Beta- Agonist

Dosing: Adult

Dosage guidance:

Dosing: Dosages are expressed as norepinephrine base; globally, product expression of dosing may vary (Ref).

Hepatorenal syndrome type 1 or acute kidney injury, treatment

Hepatorenal syndrome type 1 or acute kidney injury, treatment (alternative agent) (off-label use): **Note:** Alternative to terlipressin. Use in combination with albumin (Ref).

Continuous infusion :

Non-weight-based dosing: IV: Initial: 5 to 8 mcg/minute; dose may be increased every 4 hours based on clinical end points (eg, increase mean arterial pressure [MAP] of ~10 mm Hg from baseline, improved urine output); maximum dose: 10 mcg/minute non-ICU; 50 mcg/minute in ICU (Ref).

Hypotension or shock

Hypotension or shock:

Note: May be used in distributive, obstructive, undifferentiated, and combined shock when clinically indicated. Use in hypovolemic shock should typically be avoided (Ref). Evidence to support an

optimal dose and duration is lacking; recommendations provided are general guidelines only. Institutional protocols may vary with weight-based or non-weight-based dose regimens (Ref). Optimal goal of therapy not well established, but typically titrate to maintain end-organ perfusion.

Usual dose range:

Continuous infusion:

Weight-based dosing:

IV: Initial: 0.05 to 0.15 mcg/**kg**/minute; titrate to clinical end point (eg, MAP, organ perfusion); usual dose range: 0.025 to 1 mcg/**kg**/minute; rarely, for refractory shock, may go up to 1 to ~3 mcg/**kg**/minute (Ref).

Non-weight-based dosing:

IV: Initial: 5 to 15 mcg/minute; titrate to clinical end point (eg, MAP, organ perfusion); usual dose range: 2 to 80 mcg/minute; rarely, for refractory shock, may go up to 80 to 250 mcg/minute (Ref).

Cardiogenic shock:

Continuous infusion:

Weight-based dosing:

IV: Initial: 0.05 mcg/**kg**/minute; titrate to clinical end point (eg, MAP, organ perfusion); usual dosage range: 0.05 to 0.4 mcg/**kg**/minute (Ref).

Non-weight-based dosing:

IV: Initial: 5 mcg/minute; titrate to clinical end point (eg, MAP, organ perfusion); usual dosage range: 5 to 30 mcg/minute (Ref).

Post-cardiac arrest shock: Note: Typically titrate to maintain goal MAP of 65 mm Hg and systolic BP of 80 to 100 mm Hg to optimize cerebral and end-organ perfusion (Ref).

Continuous infusion:

Weight-based dosing:

IV: Initial: 0.05 to 0.15 mcg/**kg**/minute; titrate to clinical end point (eg, MAP, organ perfusion); usual dosing range: 0.01 to 1 mcg/**kg**/minute (Ref).

Non-weight-based dosing:

IV: Initial: 5 to 15 mcg/minute; titrate to clinical end point (eg, MAP, organ perfusion); usual dosing range: 0.5 to 80 mcg/minute (Ref).

Septic shock: Note: Typically titrate to maintain goal MAP (eg, ~65 mm Hg) (Ref).

Continuous infusion:

Weight-based dosing:

IV: Initial: 0.05 to 0.15 mcg/**kg**/minute; titrate to goal clinical endpoint (eg, BP or other measure of perfusion); usual dose range: 0.025 to 1 mcg/**kg**/minute (Ref).

Non-weight-based dosing:

IV: Initial: 5 to 15 mcg/minute; titrate to goal clinical endpoint (eg, BP or other measure of perfusion); usual dose range: 2 to 80 mcg/minute (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: No dosage adjustment necessary for any degree of kidney impairment (only small amount excreted by kidney unchanged) (Ref).

Augmented renal clearance (measured urinary CrCl ≥ 130 mL/minute/1.73 m²):

Note: Augmented renal clearance (ARC) is a condition that occurs in certain critically ill patients without organ dysfunction and with normal serum creatinine concentrations. Younger patients (<55 years of age) admitted post trauma or major surgery are at highest risk for ARC, as well as those with sepsis, burns, or hematologic malignancies. An 8- to 24-hour measured urinary CrCl is necessary to identify these patients (Ref).

No dosage adjustment necessary (Ref).

Hemodialysis, intermittent (thrice weekly): Likely to be dialyzable (Ref): No supplemental dose or dosage adjustment necessary since norepinephrine is titrated to clinical targets (Ref).

Peritoneal dialysis: No dosage adjustment necessary (Ref).

CRRT: Unlikely to be significantly dialyzable (Ref): No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, and 3 obesity (BMI ≥ 30 kg/m²):

Continuous infusion: IV: If institution uses weight-based dosing, use **ideal body weight** for initial weight-based dose calculations, then titrate to hemodynamic effect and clinical response (Ref). If institution uses nonweight-based dosing for vasoactive agents, continue with this approach. During therapy, clinicians should **not** change dosing weight from one weight metric to another (ie, ideal body weight to/from actual body weight or weight-based dosing to/from nonweight-based dosing) (Ref). Refer to adult dosing for indication-specific doses.

Rationale for recommendations: There is a paucity of studies evaluating the influence of obesity on norepinephrine dosing or pharmacokinetics. Observational studies, including norepinephrine, phenylephrine, and epinephrine, suggest nonweight-based dosing strategies may result in lower overall cumulative dose requirements and increased drug exposure to second-line agents in some patients and may not be advantageous in time to achieving hemodynamic stability (Ref). However, it is difficult to show outcome differences between weight-based and nonweight-based dosing because of dose titration to target BP, particularly in the context of retrospective studies. Furthermore, there is substantial variability in response in critically ill patients, irrespective of weight. Due to the short onset of action and small V_d, rapid titration to clinical effect after initial dosing is possible (Ref).

Dosing: Older Adult

Refer to adult dosing. For septic shock, consider initial MAP target of 60 to 65 mm Hg (Ref).

Dosing: Pediatric

(For additional information see "Norepinephrine (noraadrenaline): Pediatric drug information")

Dosage guidance:

Dosing: Dose stated in terms of **norepinephrine base**.

Hypotension/Shock, fluid resistant

Hypotension/Shock, fluid resistant: Infants, Children, and Adolescents: Continuous IV or intraosseous infusion: Initial: 0.05 to 0.1 mcg/kg/minute; titrate to desired effect; usual maximum dose: 2 mcg/kg/minute (Ref); higher doses have been reported in the literature (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Bradycardia, cardiac arrhythmia, cardiomyopathy (stress), peripheral vascular insufficiency

Central nervous system: Anxiety, transient headache

Respiratory: Dyspnea

<1%, postmarketing, and/or case reports: Peripheral gangrene, peripheral ischemia (digital [Daroca-Pérez 2017])

Contraindications

There are no contraindications listed in the US manufacturer's labeling.

Canadian labeling: Hypotension from hypovolemia except as an emergency measure to maintain coronary and cerebral perfusion until volume could be replaced; mesenteric or peripheral vascular thrombosis unless it is a lifesaving procedure; during anesthesia with cyclopropane or halothane anesthesia.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

- Extravasation: Vesicant; ensure proper needle or catheter placement prior to and during infusion. Avoid extravasation; infuse into a large vein if possible. Avoid infusion into leg veins. Monitor IV site closely. If extravasation occurs, infiltrate the area with diluted phentolamine (5 to 10 mg in 10 mL saline in adult patients) with a fine hypodermic needle. Phentolamine should be administered as soon as possible after extravasation is noted to prevent sloughing/necrosis (Stefanos 2023).

Disease-related concerns:

- Hypovolemia: Address hypovolemia before initiating therapy; patients who are hypotensive from hypovolemia may experience severe peripheral and visceral vasoconstriction, decreased renal perfusion and reduced urine output, tissue hypoxia, lactic acidosis, and reduced systemic blood flow despite normal BP.
- Hypoxia/hypercarbia: Use in patients with profound hypoxia or hypercarbia may produce ventricular tachycardia or fibrillation; use with extreme caution.

Dosage form specific issues:

- Sodium metabisulfite: Product may contain sodium metabisulfite. Use caution in patients with asthma or a sulfite allergy; allergic-type reactions, including anaphylactic symptoms and life-threatening asthmatic episodes, may occur.

Other warnings/precautions:

- Abrupt discontinuation: Gradually reduce infusion rate while expanding blood volume with IV fluids during discontinuation of therapy; severe hypotension may occur with abrupt discontinuation.
- Appropriate use: Assure adequate circulatory volume to minimize need for vasoconstrictors. Avoid hypertension; monitor BP closely and adjust infusion rate. Avoid in patients with mesenteric or peripheral vascular thrombosis; use may increase ischemia and extend the area of infarction.

Dosage Forms Considerations

Norepinephrine dosage is stated in terms of norepinephrine base. Although the IV product vial designates the contents as norepinephrine bitartrate, the actual concentration shown is in terms of norepinephrine base 1 mg/mL.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Levophed: 1 mg/mL (4 mL) [contains sodium metabisulfite]

Generic: 1 mg/mL (4 mL); 16 mg/250 mL in NaCl 0.9% (250 mL); 4 mg/250 mL in NaCl 0.9% (250 mL); 8 mg/250 mL in NaCl 0.9% (250 mL)

Solution, Intravenous [preservative free]:

Levophed: 1 mg/mL (4 mL) [contains sodium metabisulfite]

Generic: 1 mg/mL (4 mL); 16 mg/250 mL in Dextrose 5% (250 mL); 16 mg/250 mL in NaCl 0.9% (250 mL); 32 mg/250 mL in NaCl 0.9% (250 mL); 4 mg/250 mL in Dextrose 5% (250 mL); 4 mg/250 mL in NaCl 0.9% (250 mL); 8 mg/250 mL in Dextrose 5% (250 mL); 8 mg/250 mL in NaCl 0.9% (250 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Levophed Intravenous)

1 mg/mL (per mL): \$0.87

Solution (Norepinephrine Bitartrate Intravenous)

1 mg/mL (per mL): \$0.99 - \$5.69

Solution (Norepinephrine-Dextrose Intravenous)

4MG/250ML 5% (per mL): \$0.11

8MG/250ML 5% (per mL): \$0.16

16MG/250ML 5% (per mL): \$0.32

Solution (Norepinephrine-Sodium Chloride Intravenous)

4MG/250ML 0.9% (per mL): \$0.13 - \$0.17

8MG/250ML 0.9% (per mL): \$0.18 - \$0.22

16MG/250ML 0.9% (per mL): \$0.31 - \$0.36

32MG/250ML 0.9% (per mL): \$0.48

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous:

Levophea: 1 mg/mL ([DSC]) [contains sodium metabisulfite]

Generic: 1 mg/mL (4 mL); 4 mg/250 mL in Dextrose 5% (250 mL); 8 mg/250 mL in Dextrose 5% (250 mL)

Administration: Adult

IV: Administer as a continuous infusion via an infusion pump. Dilute vials prior to use; pre-mixed solutions are also available for IV infusion. Central line administration is preferred; extravasation may cause severe ischemic necrosis. If central line is not available, may administer for a short duration (<72 hours) through a peripheral IV catheter placed in a large vein at a proximal site (eg, in or proximal to antecubital fossa). Frequent monitoring of the IV catheter site is recommended to rapidly identify extravasation (Ref). Refer to institutional policies and procedures; catheter placement/size and vasopressor concentration may vary depending on institution. Administration through midline catheters may also be an option (Ref). Do not administer sodium bicarbonate (or any alkaline solution) through an IV line containing norepinephrine; inactivation of norepinephrine may occur. Avoid abrupt withdrawal; reduce infusion flow rate slowly.

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation.

Extravasation management: If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry warm compresses; initiate phentolamine (or alternative) antidote (Ref).

Phentolamine: **SUBQ:** Dilute 5 to 10 mg in 10 mL 0.9% sodium chloride and administer into extravasation site as soon as possible after extravasation; if infiltrated catheter has not been removed, administer initial dose through infiltrated catheter; may repeat in 60 minutes if patient remains symptomatic (Ref).

Alternatives to phentolamine:

Nitroglycerin topical 2% ointment: Apply a 1-inch strip to the site of ischemia to cover the affected area; may repeat every 8 hours as necessary (Ref).

Terbutaline:

Large extravasations: **SUBQ:** Infiltrate affected extravasation area with terbutaline 1 mg using a solution of terbutaline 1 mg diluted in 10 mL 0.9% sodium chloride; may repeat dose after 15 minutes (Ref).

Small extravasations: **SUBQ:** Infiltrate affected extravasation area with terbutaline 0.5 mg using a solution of terbutaline 1 mg diluted in 1 mL 0.9% sodium chloride; may repeat dose after 15 minutes (Ref).

Preparation for Administration: Adult

Continuous IV infusion:

Note: A 4 mg per 250 mL, 8 mg per 250 mL, 16 mg per 250 mL, and 32 mg per 250 mL premixed solution are also available for IV infusion.

Dilute vials with D5W, D5NS, or NS; dilution in NS is not recommended by the manufacturer; however, stability in NS has been demonstrated (Ref). Concentrations ranging from 4 to 64 mcg/mL may be used in clinical practice (Ref).

Usual Infusion Concentrations: Adult

Note: Premixed solutions are available.

IV infusion: 4 mg in 250 mL (concentration: 16 **mcg/mL**) **or** 8 mg in 250 mL (concentration: 32 **mcg/mL**) **or** 16 mg in 250 mL (concentration: 64 **mcg/mL**) **or** 32 mg in 250 mL (concentration: 128 **mcg/mL**) of D5W or NS.

Administration: Pediatric

Parenteral: Continuous IV infusion: Administer as a continuous infusion via an infusion pump. Vials must be diluted prior to administration; premixed IV solutions (16 mcg/mL and 32 mcg/mL) are available. Central line administration is preferred; extravasation may cause severe ischemic necrosis. If central line is not available, may administer for a short duration through a peripheral IV catheter placed in a large vein or via intraosseous access using a more dilute solution or with a second carrier fluid (Ref); frequent monitoring of the IV catheter site is recommended to rapidly identify extravasation (Ref). **Note:** Refer to institutional policies and procedures; catheter placement/size and vasopressor concentration may vary depending on institution. Do not administer sodium bicarbonate (or any alkaline solution) through an IV line containing norepinephrine; inactivation of norepinephrine may occur (Ref). Avoid abrupt withdrawal; reduce infusion flow rate slowly.

Rate of infusion (mL/hour) = dose (mcg/kg/minute) x weight (kg) x 60 minutes/hour divided by the concentration (mcg/mL)

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation. If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula unless needed for IV antidote; elevate extremity; apply dry, warm compresses. Initiate phentolamine (or alternative antidote) (Ref).

Preparation for Administration: Pediatric

Parenteral: Continuous IV infusion: Vial: Dilute with D5W, D5NS, or NS; dilution in NS is not recommended by the manufacturer; however, stability in NS has been demonstrated (Ref). Vermont Oxford Network recommends a standard concentration of 4 mcg/mL or 16 mcg/mL for neonates (Ref). For pediatric patients, ASHP recommends standard concentrations of 16 mcg/mL, 32 mcg/mL, and 64 mcg/mL (Ref).

Usual Infusion Concentrations: Pediatric

Note: Premixed solutions are available.

IV infusion: 16 **mcg/mL**, 32 **mcg/mL**, **or** 64 **mcg/mL**.

Vesicant/Extravasation Risk

Vesicant

Storage/Stability

Premixed solution in D5W: Store in protective overwrap at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from light. Once overwrap is removed, may be stored for up to 30 days at room temperature.

Premixed solution in NS: Store in protective overwrap at 20°C to 25°C (68°F to 77°F). Protect from light. Once overwrap is removed, may store at room temperature for up to 7 to 45 days (varies by manufacturer; refer to product labeling).

Vials: Store intact vials in original container at 20°C to 25°C (68°F to 77°F); excursion permitted to 15°C to 30°C (59°F to 86°F). Diluted solution may be stored at room temperature for up to 24 hours. Protect from light.

Use: Labeled Indications

Hypotension or shock: Treatment of severe hypotension, cardiogenic shock, or septic shock (and other vasodilatory shock states) that persist after adequate fluid volume replacement.

Use: Off-Label: Adult

Hepatorenal syndrome type 1 or acute kidney injury, treatment

Medication Safety Issues

Sound-alike/look-alike issues:

Levophed may be confused with levofloxacin

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (adrenergic agonist, IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

Substrate of COMT

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Alpha1-Blockers: May decrease therapeutic effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Antidiabetic Agents: Hyperglycemia-Associated Agents may decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Atomoxetine: May increase hypertensive effects of Sympathomimetics. Atomoxetine may increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Azosemide: May decrease therapeutic effects of Norepinephrine. *Risk C: Monitor*

Benzylpenicilloyl Polylysine: Coadministration of Alpha-/Beta-Agonists and Benzylpenicilloyl Polylysine may alter diagnostic results. Management: Consider delaying skin testing until alpha/beta agonists are no longer required and their effects have diminished, or use a histamine skin test as a positive control to assess a patient's ability to mount a wheal and flare response. *Risk D: Consider Therapy Modification*

Bitter Orange: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Bornaprine: Sympathomimetics may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Bromocriptine: May increase hypertensive effects of Alpha-/Beta-Agonists. Management: Consider alternatives to this combination when possible. If combined, monitor for hypertension and tachycardia, and do not coadminister these agents for more than 10 days. *Risk D: Consider Therapy Modification*

Cannabinoid-Containing Products: May increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Chloroprocaine (Systemic): May increase hypertensive effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

CloZAPine: May decrease therapeutic effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Cocaine (Topical): May increase hypertensive effects of Sympathomimetics. Management: Consider alternatives to use of this combination when possible. Monitor closely for substantially increased blood pressure or heart rate and for any evidence of myocardial ischemia with concurrent use. *Risk D: Consider Therapy Modification*

COMT Inhibitors: May increase serum concentrations of COMT Substrates. *Risk C: Monitor*

Dihydralazine: Sympathomimetics may decrease therapeutic effects of Dihydralazine. *Risk C: Monitor*

Doxapram: Sympathomimetics may increase adverse/toxic effects of Doxapram. *Risk C: Monitor*

Doxofylline: Sympathomimetics may increase adverse/toxic effects of Doxofylline. *Risk C: Monitor*

Droxidopa: Sympathomimetics may increase hypertensive effects of Droxidopa. *Risk C: Monitor*

Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates): May increase vasoconstricting effects of Alpha-/Beta-Agonists. *Risk X: Avoid*

Esketamine (Injection): May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for elevated heart rate, hypertension, and arrhythmias may be increased. *Risk C: Monitor*

Furosemide: May decrease therapeutic effects of Norepinephrine. *Risk C: Monitor*

Guanethidine: May increase hypertensive effects of Sympathomimetics. Guanethidine may increase arrhythmogenic effects of Sympathomimetics. *Risk C: Monitor*

Hexoprenaline: May increase adverse/toxic effects of Alpha-/Beta-Agonists. *Risk X: Avoid*

Hyaluronidase: May increase vasoconstricting effects of Alpha-/Beta-Agonists. Management: Do not use hyaluronidase to enhance the dispersion or absorption of alpha-/beta-agonists. Use of hyaluronidase for other purposes in patients receiving alpha-/beta-agonists may be considered as clinically indicated. *Risk D: Consider Therapy Modification*

Inhalational Anesthetics: May increase arrhythmogenic effects of Norepinephrine. *Risk C: Monitor*

Kratom: May increase adverse/toxic effects of Sympathomimetics. *Risk X: Avoid*

Landiolol: Sympathomimetics may decrease therapeutic effects of Landiolol. *Risk C: Monitor*

Linezolid: May increase hypertensive effects of Sympathomimetics. Management: Consider initial dose reductions of sympathomimetic agents, and closely monitor for enhanced blood pressure elevations, in patients receiving linezolid. *Risk D: Consider Therapy Modification*

Lisuride: May increase hypertensive effects of Alpha-/Beta-Agonists. Management: Consider alternatives when possible, as product labeling states the concomitant use of lisuride with alpha-/beta-agonists is not recommended. If combined, monitor for increases in blood pressure or hypertensive crises. *Risk D: Consider Therapy Modification*

Monoamine Oxidase Inhibitors: May increase hypertensive effects of Norepinephrine. *Risk C: Monitor*

Pergolide: May increase hypertensive effects of Alpha-/Beta-Agonists. *Risk C: Monitor*

Serotonin/Norepinephrine Reuptake Inhibitor: May increase tachycardic effects of Alpha-/Beta-Agonists. Serotonin/Norepinephrine Reuptake Inhibitor may increase vasopressor effects of Alpha-/Beta-Agonists. Management: If possible, avoid coadministration of direct-acting alpha-/beta-agonists and serotonin/norepinephrine reuptake inhibitors. If coadministered, monitor for increased sympathomimetic effects (eg, increased blood pressure, chest pain, headache). *Risk D: Consider Therapy Modification*

Solriamfetol: Sympathomimetics may increase hypertensive effects of Solriamfetol. Sympathomimetics may increase tachycardic effects of Solriamfetol. *Risk C: Monitor*

Sympathomimetics: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Tedizolid: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for increased blood pressure and heart rate may be increased. *Risk C: Monitor*

Thyroid Products: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk of hypertension, tachycardia, and coronary insufficiency may be increased in patients with coronary artery disease. *Risk C: Monitor*

Tricyclic Antidepressants: May increase vasopressor effects of Alpha-/Beta-Agonists. Management: Avoid, if possible, the use of alpha-/beta-agonists in patients receiving tricyclic antidepressants. If combined, monitor for evidence of increased pressor effects and consider reductions in initial dosages of the alpha-/beta-agonist. *Risk D: Consider Therapy Modification*

Vasopressin: Alpha-/Beta-Agonists (Direct-Acting) may increase hypertensive effects of Vasopressin. The effect of other hemodynamic parameters may also be enhanced. *Risk C: Monitor*

Pregnancy Considerations

Norepinephrine is an endogenous catecholamine and crosses the placenta (Minzter 2010; Wang 1999).

Medications used for the treatment of cardiac arrest in pregnancy are the same as in the non-pregnant woman. Appropriate medications should not be withheld due to concerns of fetal teratogenicity. Norepinephrine use during the post-resuscitation phase may be considered; however, the effects of vasoactive medications on the fetus should also be considered. Doses and indications should follow current Advanced Cardiovascular Life Support guidelines (Jeejeebhoy [AHA] 2015).

Breastfeeding Considerations

It is not known if norepinephrine is present in breast milk. The manufacturer recommends that caution be exercised when administering norepinephrine to breastfeeding women.

Monitoring Parameters

Hemodynamics (eg, BP, heart rate, systemic vascular resistance), end-organ function, peripheral perfusion.

Consult individual institutional policies and procedures.

Mechanism of Action

Stimulates beta₁-adrenergic receptors and alpha-adrenergic receptors causing increased contractility and heart rate as well as vasoconstriction, thereby increasing systemic blood pressure and coronary blood flow; clinically, alpha effects (vasoconstriction) are greater than beta effects (inotropic and chronotropic effects)

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Very rapid acting.

Duration: Vasopressor: 1 to 2 minutes.

Distribution: V_d: 8.8 L.

Protein binding: 25% mainly albumin with smaller extent to prealbumin and alpha 1-acid glycoprotein.

Metabolism: Via catechol-o-methyltransferase and monoamine oxidase.

Half-life elimination: Mean: ~2.4 minutes.

Time to peak, serum: Steady state: 5 minutes.

Excretion: Urine (as inactive metabolites; small amounts as unchanged drug).

Patient Counseling Points

(For additional information see "Norepinephrine (noraðrenaline): Patient drug information")

What is this drug used for?

- It is used to treat low blood pressure.

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Severe headache
- Dizziness
- Passing out
- Vision changes
- Slow heartbeat
- Abnormal heartbeat
- Anxiety
- Shortness of breath
- Excessive weight gain
- Swelling of arms or legs
- Severe injection site pain, swelling, burning, or irritation
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Levophea | Noraðrenaline;
- (AR) Argentina: Noradrenalina biol | Noradrenalina phoroneus | Noradrenalina richet | Norepinefrina gray | Norepinefrina northia;
- (AT) Austria: Noradrenalin Aguetant | Noradrenalin hospira | Noradrenalin kabi | Noradrenalin mayrhofer | Noradrenalin orpha | Norepinephrin kalceks | Sinora;

- (AU) Australia: Levophea | Noraadrenaline bnm | Noraadrenaline junio | Noraadrenaline myx | Noralin;
- (BD) Bangladesh: Hyponor | Levofed | Maxnor;
- (BE) Belgium: Levophed | Noraadrenaline (norepinephrine) kalceks | Noraadrenaline aguettant | Norepine;
- (BG) Bulgaria: Noradrenalin orpha | Norepinephrine sopharma | Sinora | Stenor pph;
- (BR) Brazil: Epikabi | Hemitartarato de Norepinefrina | Hyponor | Levophea | Noradren | Nora-
- (CH) Switzerland: Noraadrenaline sintetica;
- (CL) Chile: Cardolina | Norepinefrina | Pridam;
- (CO) Colombia: ADS-Noltron | Levophea | Norepinefrina | Pridam;
- (CR) Costa Rica: Biemefrin | Norepinefrina | Softamicid;
- (CZ) Czech Republic: Noradrenalin | Norepinephrine kabi | Norepinephrine kalceks | Sinora;
- (DE) Germany: Noradrenalin Aguettant | Noradrenalin kabi | Norepinephrin kalceks | Sinora;
- (EC) Ecuador: Norepinefrina | Norepinefrina bitartrato | Norepinefrina vitalis;
- (EE) Estonia: Biemefrin | Cardenor | Noradren | Noradrenalin | Noraadrenaline | Norepine | Norepinephrine kabi | Norepinephrine kalceks | Norepinephrine sintetica | Norepinephrine sopharma | Sinora;
- (EG) Egypt: Levophrine;
- (ES) Spain: Noradrenalina Braun | Noradrenalina Combino pharm | Noradrenalina kalceks | Noradrenalina Normon | Norages;
- (FI) Finland: Noradrenalin | Noradrenalin abcur | Noradrenalin Aguettant | Noradrenalin fresenius kabi | Noradrenalin hospira | Noradrenalin kalceks | Noraadrenaline sintetica;
- (FR) France: Cronomir | Noraadrenaline aguettant | Noraadrenaline Hospira | Noraadrenaline Merck | Noraadrenaline PCH | Noraadrenaline renaudin | Noraadrenaline tartrate kalceks;
- (GB) United Kingdom: Levophea | Noraadrenaline | Sinora;
- (GR) Greece: Noradren;
- (HK) Hong Kong: Levophed | Noraadrenaline sintetica;
- (HR) Croatia: Noradrenalin ligula pharma;
- (HU) Hungary: Noraadrenaline kabi | Norepinephrine kalceks | Sinora;
- (ID) Indonesia: Epinor | Epivas | Guprin | Levophea | Norepinephrine | Raivas | Vascon;
- (IE) Ireland: Levophed | Noraadrenaline | Noraadrenaline (Norepinephrine);
- (IN) India: Adrenor | Akunor ad | Infunor | Norad | Noradol | Noradria | Norepirin;
- (IT) Italy: Noradrenalina tartrato | Noradrenalina tartrato aguettant;
- (JO) Jordan: Levophed | Noraadrenaline;
- (JP) Japan: Levophed | Noradrenalin;
- (KR) Korea, Republic of: Levophed | Q phrine;
- (KW) Kuwait: Levophea | Noraadrenaline | Noraadrenaline aguettant;
- (LB) Lebanon: Noraadrenaline tartrate renaudin;
- (LT) Lithuania: Levonor | Levophed | Norad 4 | Noradren | Noraadrenaline (tartrate) aguettant | Norages | Norepine | Norepinephrine kalceks | Norepinephrine sintetica | Norepinephrine sopharma;
- (LU) Luxembourg: Levophea;
- (LV) Latvia: Norepinephrine kabi | Norepinephrine kalceks | Norepinephrine sopharma | Sinora;
- (MX) Mexico: Afhimad | Boguefren | Ednalen | Envetim | Naretia | Necsol | Nepridar | Softamicid;
- (MY) Malaysia: Biemefrin | Cardiamed | Levophea;
- (NL) Netherlands: Noraadrenaline Hospira | Noraadrenaline kabi | Noraadrenaline kalceks | Norepinefrine | Sinora;
- (NO) Norway: Noradrenalin | Noradrenalin abcur | Noradrenalin Aguettant | Noradrenalin kalceks | Noradrenalin macure | Noradrenalin sintetica;
- (NZ) New Zealand: Biomed noraadrenaline | Levophea | Noraadrenaline;
- (PA) Panama: Biemefrin | Noradrenalina | Noradrenalina richet | Norepinefrina | Pridam;
- (PE) Peru: Niacor | Norepinefrina | Norepinefrina northia | Pinephed;
- (PH) Philippines: Hypoact | Levophea | Mephrin | Norepin | Norphed | Northix | Sandine;
- (PL) Poland: Adrenor | Cardenor | Noradrenalin kalceks | Noraadrenaline aguettant | Noraadrenaline kabi | Norepinephrine | Sinora;
- (PR) Puerto Rico: Levophea | Norepinephrine;

- (PT) Portugal: L noradrenalina braun | Noradrenalina | Noradrenalina basi;
- (PY) Paraguay: Norepinefrina northia;
- (QA) Qatar: Biemefrin | Cardenor | Levophed | Noraadrenaline Aguetant | Veraline;
- (RO) Romania: Noradrenalina amdipharm | Noradrenalina kalceks;
- (RU) Russian Federation: Noraadrenaline | Noraadrenaline aguetant | Noraadrenaline kabi | Norepinephrine;
- (SA) Saudi Arabia: Epinor | Levophed | Nevoleen | Noradren;
- (SE) Sweden: Noradrenalin abcur | Noradrenalin Aguetant | Noradrenalin fresenius kabi | Noradrenalin hospira | Noradrenalin kalceks | Noraadrenaline sintetica;
- (SG) Singapore: Levophed;
- (SI) Slovenia: Noradrenalin kabi | Noradrenalin sintetica;
- (SK) Slovakia: Norepinephrine kabi | Norepinephrine kalceks | Sinora;
- (SV) El Salvador: Pridam;
- (TH) Thailand: Levophed | N epi | Noraadrenaline aguetant | Norene | Norpin;
- (TN) Tunisia: Noraadrenaline | Noraadrenaline medis | Noraadrenaline mylan | Noraadrenaline renaudin | Noraadrenaline saiph | Noraline;
- (TR) Turkey: Adrenor | Biemefrin | Cardenor | Epinor | Forefrin | Nefrinor | Noreprin | Seladrenalin | Stenor | Steradin | Turktipsan noradrenalin;
- (TW) Taiwan: Levophed bitartrat | Nobify | Noradrec | Norepine | Norepinephrine;
- (UA) Ukraine: Adrenor | N epi | Norepinephrin kalceks;
- (UG) Uganda: Norad;
- (UY) Uruguay: Fioritina | Noradrenalina biol | Noradrenalina novophar | Norepinefrina northia;
- (ZA) South Africa: Sinora

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